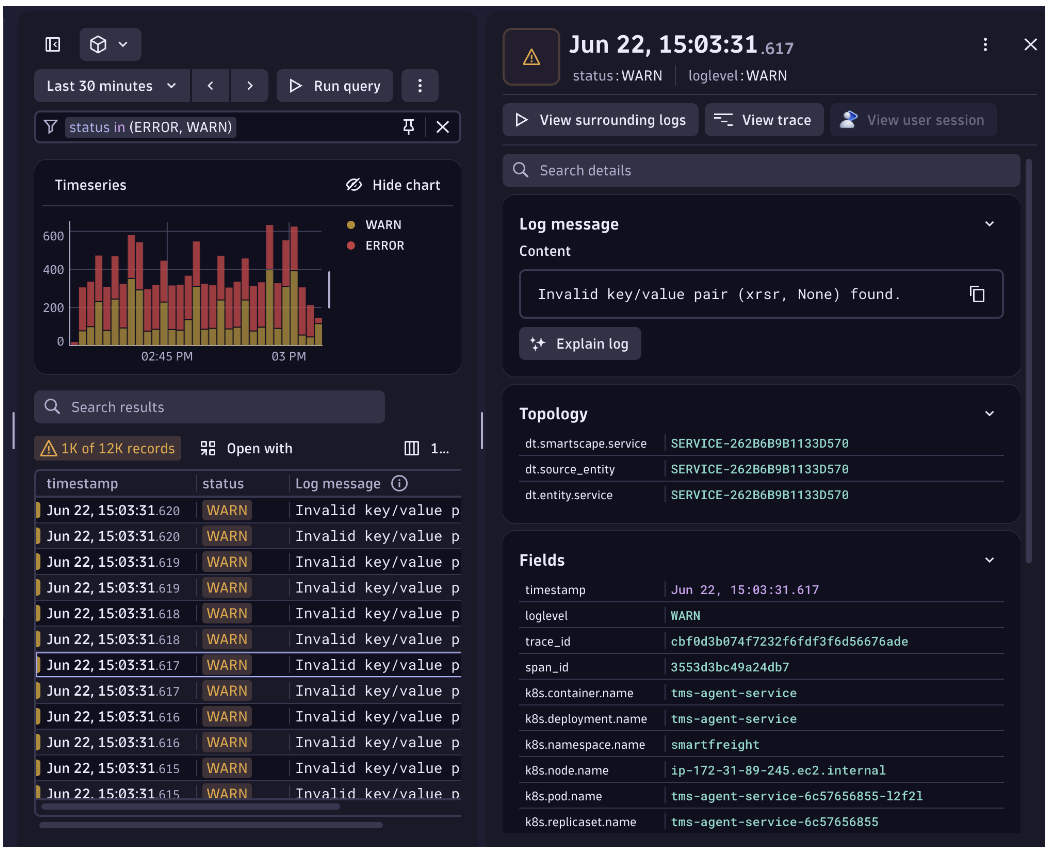
<u>Dynatrace Anomaly Detection Workflow</u>

Step 1: Data Collection(NO AI OR ML IS USED HERE)

Dynatrace collects telemetry data from across the entire technology stack using OneAgent. This includes logs, metrics, traces, events, infrastructure data, and user experience data from applications, servers, containers, databases, and cloud services.

Step 2: Data Enrichment and Preprocessing (NO AI OR ML IS USED HERE)

The collected data is automatically enriched with contextual information such as service names, host details, Kubernetes metadata, trace IDs, process information, and cloud attributes.



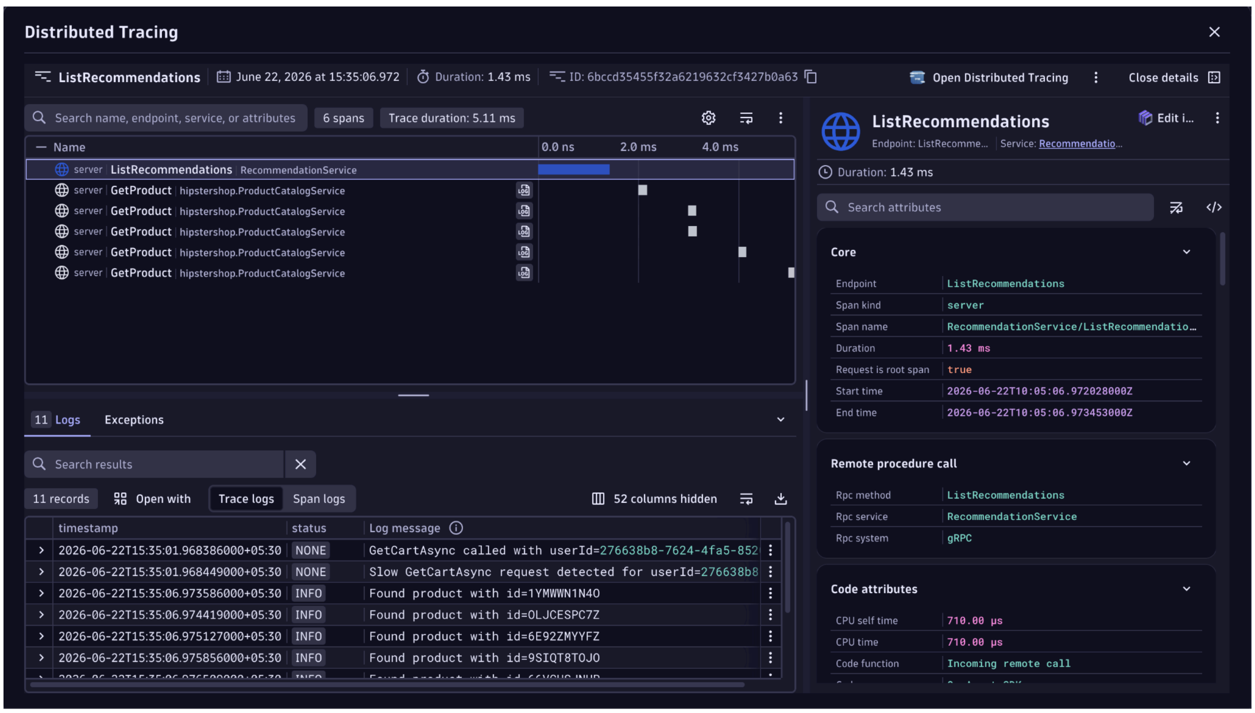
Gets data about topology and other related fields using logs

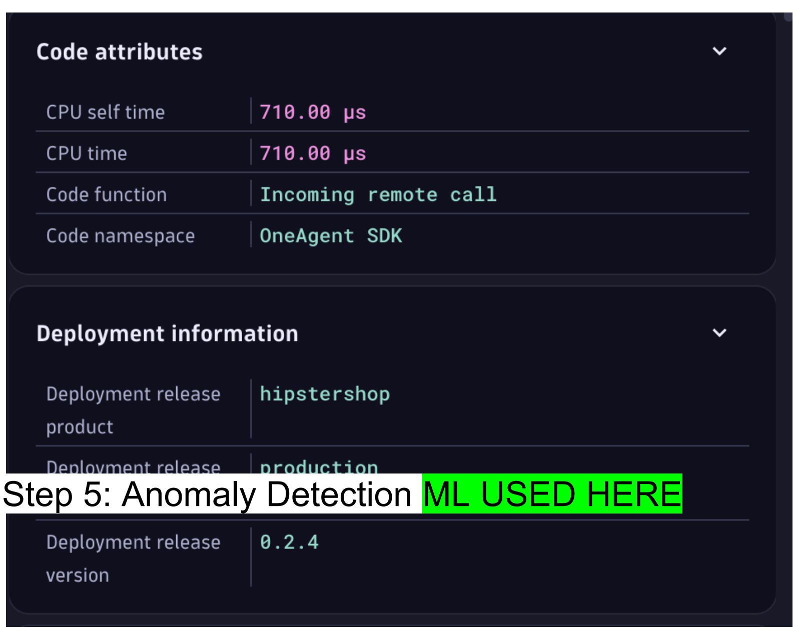
Step 3: Topology Discovery (Smartscape) (NO AI OR ML IS USED HERE)

Dynatrace automatically builds a real-time dependency map called Smartscape. It identifies relationships between applications, services, processes, hosts, containers, databases, and cloud resources, creating a complete view of the system architecture.

Step 4: Baseline Learning(This is where ML starts.)

Davis AI continuously analyzes historical data to learn normal behavior patterns for key metrics such as response time, CPU utilization, memory consumption, throughput, traffic volume, and error rates. These learned patterns become dynamic baselines.





Step 5: Anomaly Detection ML USED HERE

Incoming data is continuously compared against the learned baselines. If a metric deviates significantly from its expected behavior, Dynatrace automatically flags it as an anomaly. Examples include sudden spikes in response time, increased error rates, abnormal resource consumption, or unexpected traffic patterns.

Step 6: Correlation Analysis Graph Algorithms

Instead of treating anomalies as isolated events, Dynatrace correlates logs, metrics, traces, and events using the Smartscape dependency model. This helps determine how anomalies are connected across the system.

Step 7: Root Cause Analysis(this is where DAVIS AI is used)

Davis AI performs causal analysis to identify the actual source of the problem. For example, if a database slowdown causes failures in a payment service and API gateway, Dynatrace traces the chain of dependencies and identifies the database as the root cause.

Step 8: Impact Assessment Graph Algorithms

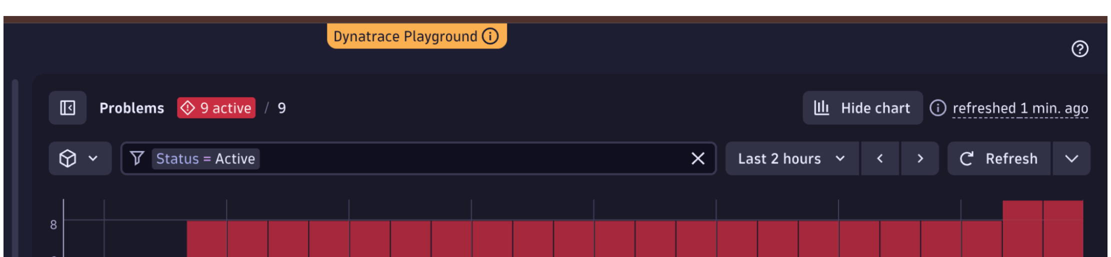
Dynatrace determines which applications, services, infrastructure components, and users are affected by the anomaly. This helps teams understand the business impact of the issue.

Step 9: Problem Creation Some ML may be involved. / correlation

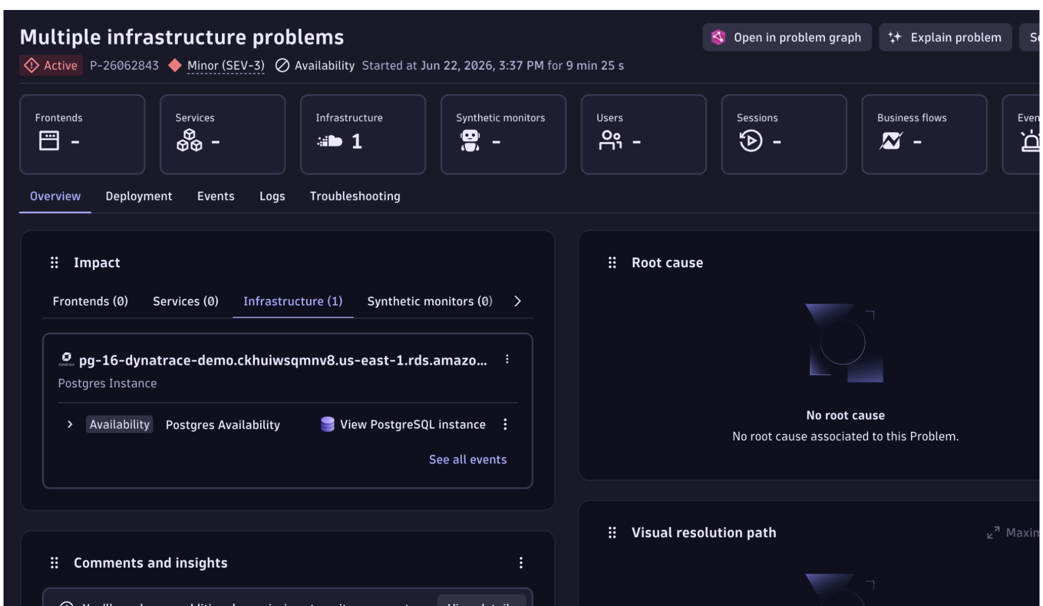
Rather than generating hundreds of separate alerts, Dynatrace groups related anomalies into a single problem card. This reduces alert noise and provides a consolidated view of the incident.

Step 10: Actionable Insights given by LLM

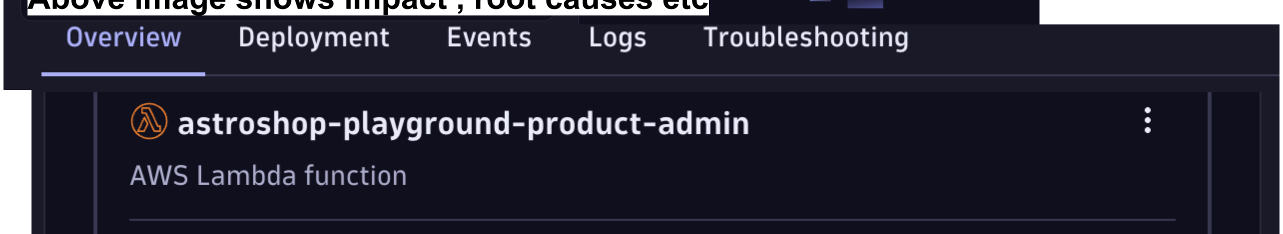
Finally, Dynatrace provides the detected anomaly, identified root cause, affected services, severity level, business impact, and recommended remediation actions, enabling faster troubleshooting and resolution.



All the problems identified by Davis AI + Detection Rules + Monitoring Rules



Above image shows impact , root causes etc



Anomaly detection in action

Dynatrace performs anomaly detection by continuously collecting metrics, logs, and traces from applications and infrastructure. Davis AI first learns the normal behavior of the system by analyzing historical patterns such as response times, error rates, CPU usage, memory consumption, and traffic volume. It then compares incoming real-time data against these learned baselines. When a metric deviates significantly from its expected behavior—for example, a sudden spike in response time or error rate—Dynatrace flags it as an anomaly. Instead of treating anomalies as isolated events, Davis AI correlates them using topology information, service dependencies, and distributed traces to understand how components are connected. This allows it to identify the most likely root cause, determine affected services, assess business impact, and generate a single problem card rather than multiple independent alerts.

How does company uses Dynatrace ?

Companies use Dynatrace by installing its OneAgent software on their servers, applications, virtual machines, containers, and cloud environments. The agent continuously collects telemetry data such as logs, metrics, traces, CPU usage, memory consumption, network activity, and application performance information. This data is automatically sent to the Dynatrace platform, where it is stored and analyzed in real time. Dynatrace's AI engine, Davis AI, learns the normal behavior of the system and establishes performance baselines. When unusual patterns such as error spikes, increased response times, service failures, or infrastructure issues occur, Davis AI automatically detects the anomaly, correlates related events across the entire application stack, identifies the most likely root cause, and generates a problem report with impacted services and recommended actions. This helps organizations monitor complex systems, reduce downtime, and troubleshoot issues much faster than traditional manual log analysis methods.

Another point to note :

When companies say "Dynatrace uses AI", it is not feeding all logs into ChatGPT-like LLMs.

The AI/ML is mainly used in anomaly detection, correlation, and root cause analysis.